

AM 234 / BME 234:
Machine Learning and Artificial Intelligence
in Genomics

Class 3: April 7, 2026

Spring Quarter 2026

Topic 1: Hidden Markov models (HMMs) and genome annotation

References

Durbin et al. *Biological Sequence Analysis*. Chapter 3.

http://www.mcb111.org/w06/durbin_book.pdf

Kellis et al. *Computational Biology - Genomes, Networks, and Evolution*.
Chapters 7 & 8.

<https://batch.libretexts.org/print/Finished/bio-40877/Full.pdf>

Short primers:

<https://www.nature.com/articles/nbt1004-1315>

<https://www.nature.com/articles/s41592-019-0532-6>

<https://www.nature.com/articles/s41592-019-0702-6>

How to analyze an unannotated sequence?

```
...TCAGACGCTTGGTCAGAGGTCGCTACGTGTAAAGGTGGGTCAAGAACCCACAGT  
ATGGACGCCGGTCACCTGAGCTTCGTGGAAGAGGTGTTTGAGAACCAGACCCGG  
ACGTTGCATGCGCTATGACCATCAAGAATAGCGGTAGCTACGATCGGGATTCTGA...
```

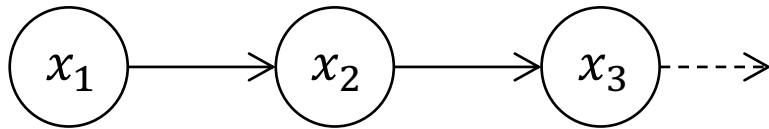
Strategies:

- Find matching sequences (sequence alignment).
- Examine nucleotide frequencies or k-mer frequencies.
- Model underlying 'states' of the sequence (e.g. exon, intron, promoter).

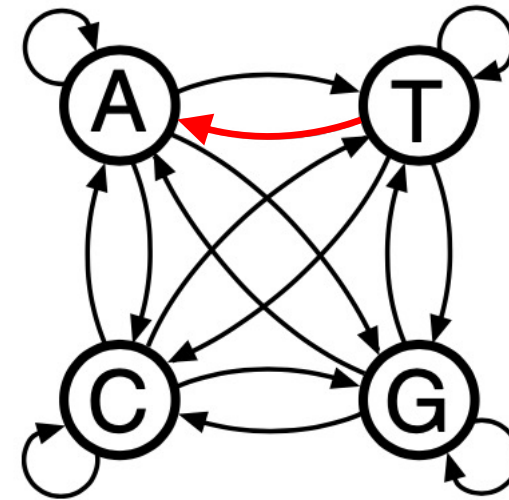
Markov chains

...TCAGACGCTTGGTCAGAGGTCGCTACGTGTAAAGGTGGGTCAAGAACCCACAGT...

Consider the DNA sequence to be a sequence of random variables, each with 4 options: A, C, G, T



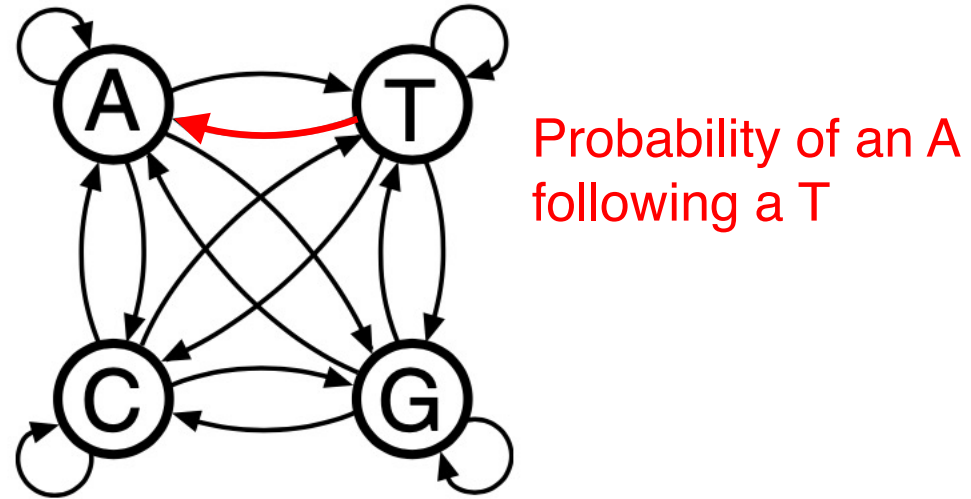
Sequence of random variables



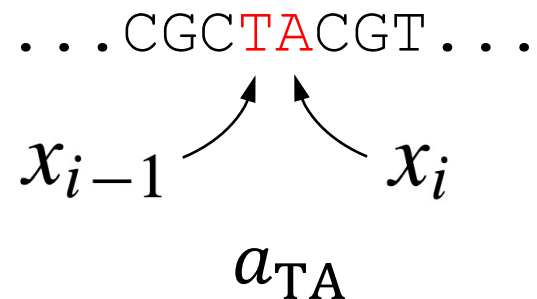
Probability of an A
following a T

Transitions between variables

Markov chains



Transition probabilities a_{st} : $a_{st} = P(x_i = t | x_{i-1} = s)$



Probability of a sequence

...TCAGACGCTTGGTCAGAGGTCGCTACGTGTAAAGGTGGGTCAAGAACCCACAGT...

$$\begin{aligned} P(x) &= P(x_L, x_{L-1}, \dots, x_1) \\ &= P(x_L | x_{L-1}, \dots, x_1) P(x_{L-1} | x_{L-2}, \dots, x_1) \cdots P(x_1) \end{aligned}$$

Chain rule

$$P(X, Y) = P(X|Y)P(Y)$$

Markov assumption:

The probability of observing a given base depends only on the previous base, not on the entire previous sequence.

Markov assumption

The probability of observing a given base depends only on the previous base, not on the entire previous sequence.

$$\begin{aligned} P(x_i | x_{i-1}, \dots, x_1) &= P(x_i | x_{i-1}) \\ &= a_{x_{i-1}x_i} \end{aligned}$$

Transition probability
from x_{i-1} to x_i

Each observation in the Markov chain is independent of all other observations when conditioned on the preceding observation.

Homogeneity: Transition probabilities are constant (do not vary over time).

Probability of a sequence

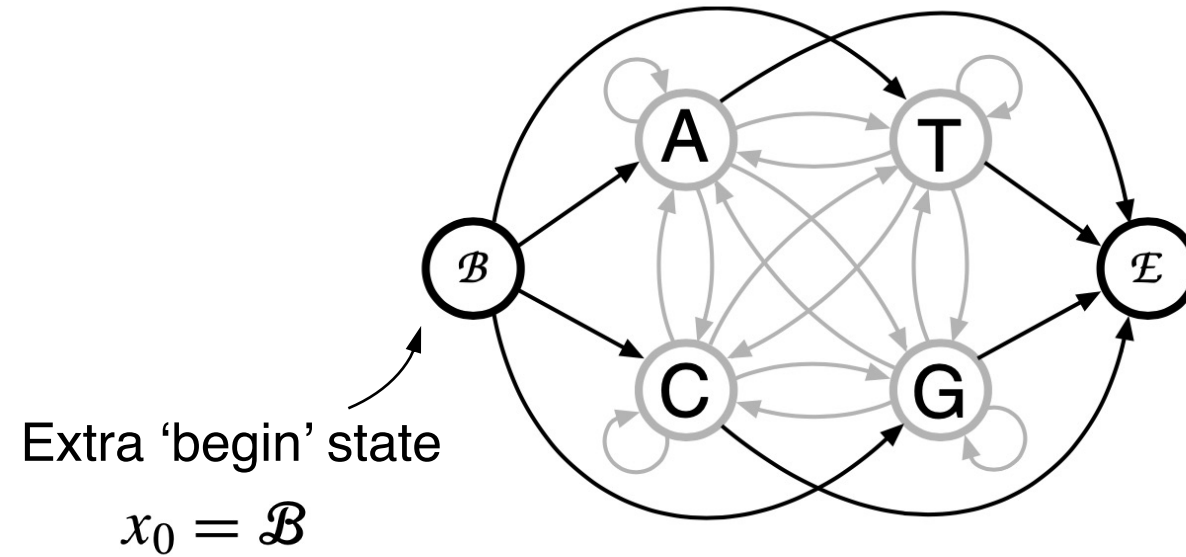
...TCAGACGCTTGGTCAGAGGTCGCTACGTGTAAAGGTGGGTCAAGAACCCACAGT...

$$\begin{aligned}P(x) &= P(x_L, x_{L-1}, \dots, x_1) \\&= P(x_L | x_{L-1}, \dots, x_1) P(x_{L-1} | x_{L-2}, \dots, x_1) \cdots P(x_1) \\&= P(x_L | x_{L-1}) P(x_{L-1} | x_{L-2}) \cdots P(x_2 | x_1) P(x_1) \\&= P(x_1) \prod_{i=2}^L a_{x_{i-1}x_i}\end{aligned}$$

Probability of starting
in state x_1

Product over all state transitions

Starting and ending probabilities



$$P(x_1 = s) = a_{\mathcal{B}s}$$

AGGTC...

$$P(x) = a_{\mathcal{B}A} * a_{AG} * a_{GG} * a_{GT} * a_{TC} * \dots$$

Example DNA sequence Markov models

CpG island model:

+	A	C	G	T
A	0.180	0.274	0.426	0.120
C	0.171	0.368	0.274	0.188
G	0.161	0.339	0.375	0.125
T	0.079	0.355	0.384	0.182

Non-CpG island model:

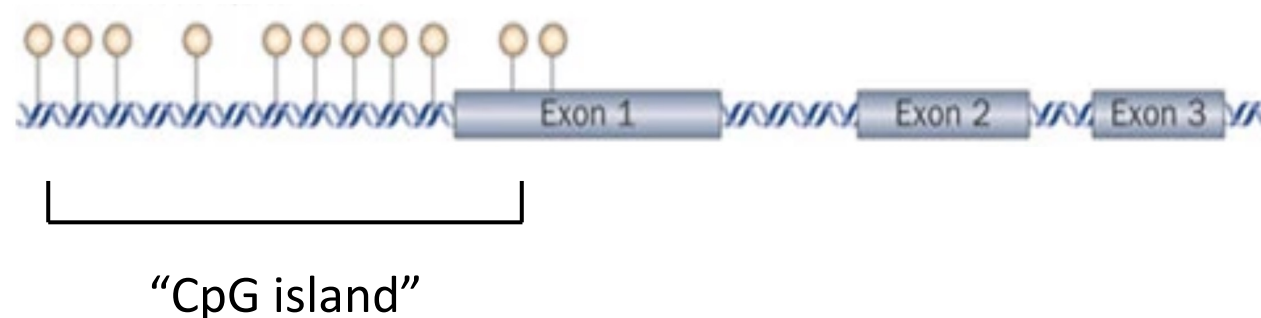
—	A	C	G	T
A	0.300	0.205	0.285	0.210
C	0.322	0.298	0.078	0.302
G	0.248	0.246	0.298	0.208
T	0.177	0.239	0.292	0.292

Example DNA sequence Markov models

DNA methylation occurs at C's followed by G's (CpG's):



CpG's are more often found near genes:



Comparing two models

...TCAGACGCTTGGTCAGAGGTCGCTACGTGTAAAGGTGGGTCAAGAACCCACAGT...

What is the probability that this sequence came from one model or the other?

Log odds ratio:
$$\log \frac{P(x|\text{model } +)}{P(x|\text{model } -)} = \sum_{i=1}^L \log \frac{a_{x_{i-1}x_i}^+}{a_{x_{i-1}x_i}^-}$$

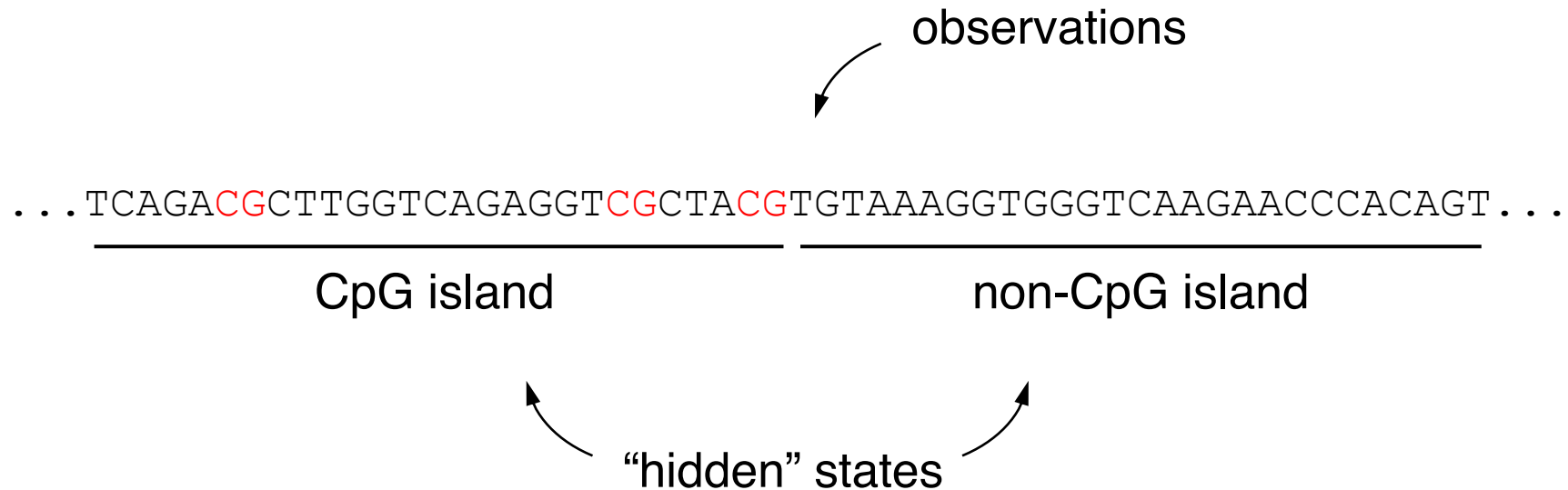
This comparison assumes the entire sequence came from one model.

How do we find different regions within a given sequence that come from one model vs another?

Switching between models

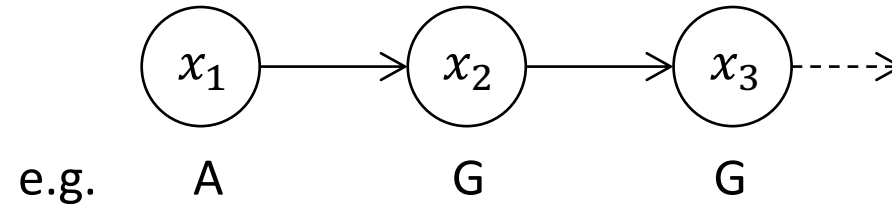
How do we find different regions within a given sequence that come from one model vs another?

e.g.

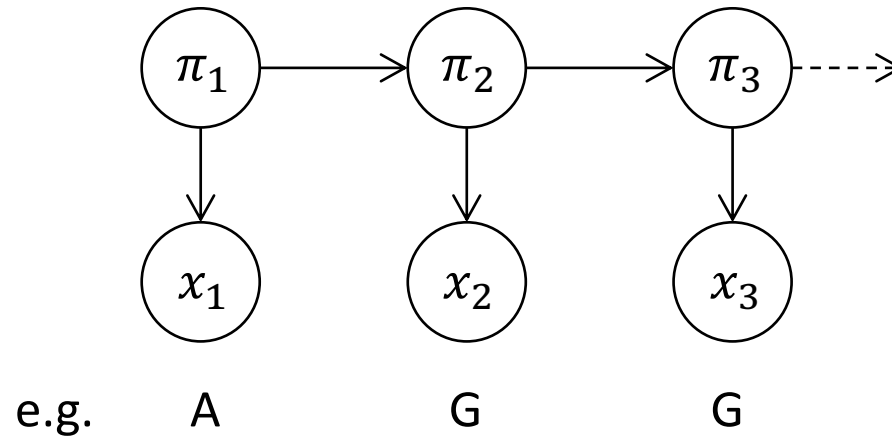


Hidden Markov model (HMM)

Previously (Markov chain):



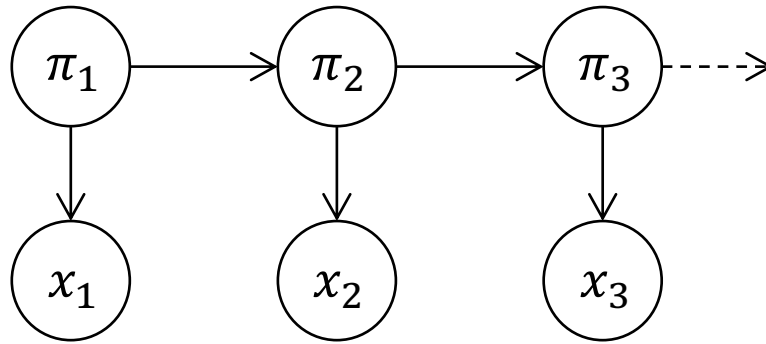
Add hidden states:



Hidden Markov model (HMM)

Hidden states follow a Markov chain
with transition probabilities:

$$a_{kl} = P(\pi_i = l | \pi_{i-1} = k)$$



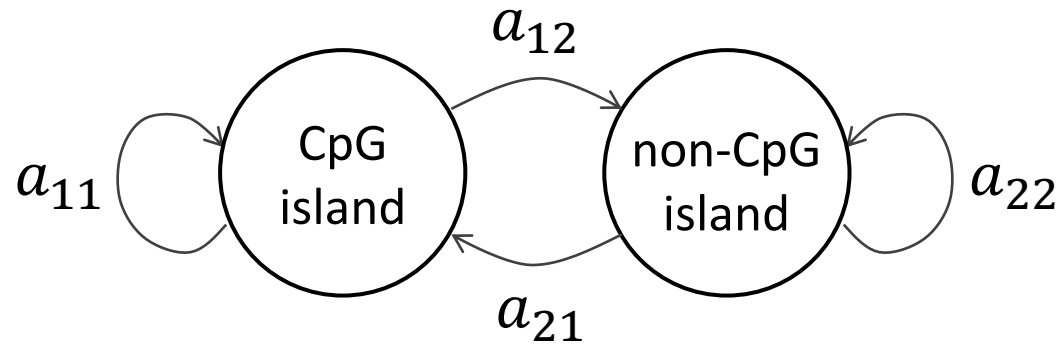
Observations (emissions) depend only
on the corresponding hidden state:

$$e_k(b) = P(x_i = b | \pi_i = k)$$

(emission probabilities)

Example HMM for finding CpG islands in DNA

Two state model:



$e_1(A)$

$e_2(A)$

$e_1(C)$

$e_2(C)$

$e_1(G)$

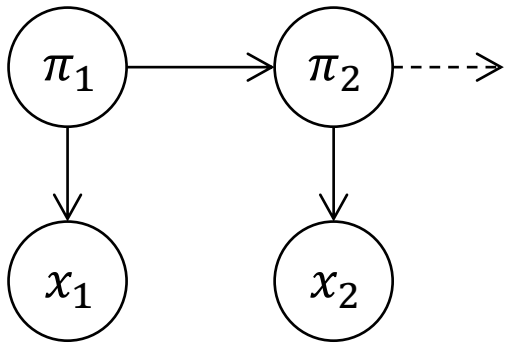
$e_2(G)$

$e_1(T)$

$e_2(T)$

Probability of a sequence

Joint probability of an observed sequence and hidden state sequence, given the model parameters:



$$P(x, \pi) = a_{0\pi_1} * e_{\pi_1}(x_1) * a_{\pi_1\pi_2} * e_{\pi_2}(x_2) * a_{\pi_2\pi_3} * \dots$$

$$= a_{0\pi_1} \prod_{i=1}^L e_{\pi_i}(x_i) a_{\pi_i\pi_{i+1}}$$

Probability of starting
in hidden state π_1

Product over all emissions and
transitions from each hidden state

If hidden states are unknown...

Decoding: Finding the most likely hidden states given the observations and model parameters.

1. Viterbi algorithm:
Finds the most likely sequence of hidden states.

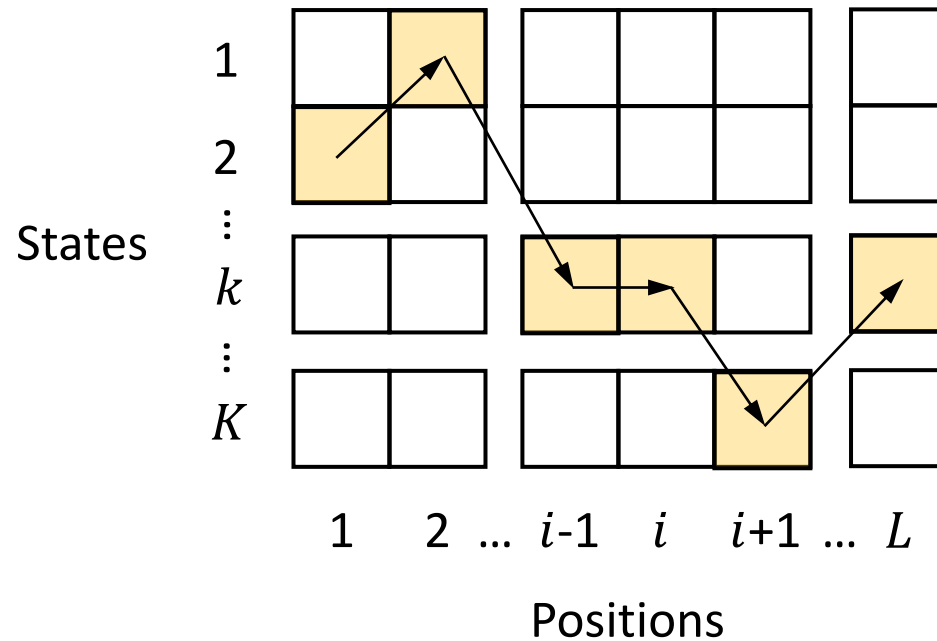
$$\pi^* = \operatorname{argmax}_{\pi} P(x, \pi)$$

2. Forward-backward algorithm (posterior decoding):
Finds the most likely hidden state at each position.

$$\hat{\pi}_i = \operatorname{argmax}_k P(\pi_i = k | x)$$

Viterbi algorithm

Find the most likely sequence of hidden states (i.e. most likely path through hidden state space):



$$\pi^* = \operatorname{argmax}_{\pi} P(x, \pi)$$

Viterbi algorithm

$$\pi^* = \operatorname{argmax}_{\pi} P(x, \pi)$$

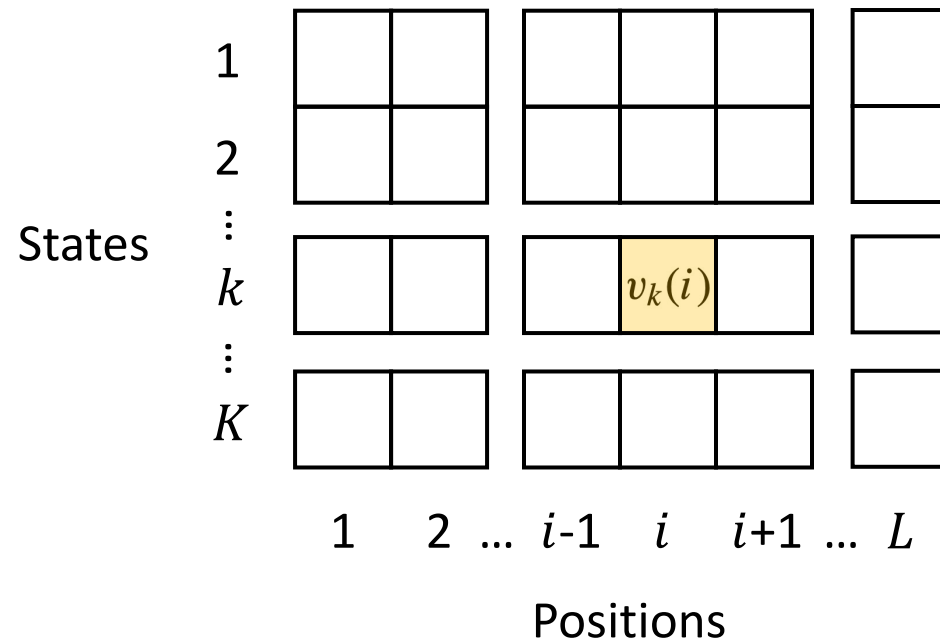
We know how to calculate $P(x, \pi)$:

$$P(x, \pi) = a_{0\pi_1} \prod_{i=1}^L e_{\pi_i}(x_i) a_{\pi_i \pi_{i+1}}$$

But enumerating all possible state sequences is infeasible.

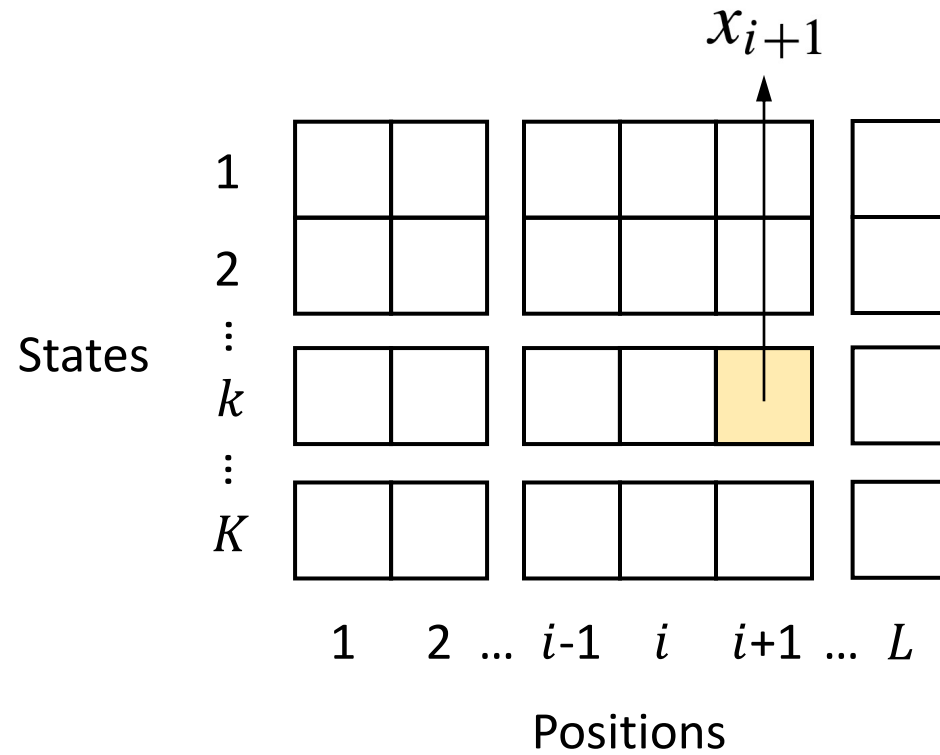
Instead, break the problem down into subproblems and solve them recursively (dynamic programming).

Viterbi algorithm



$v_k(i)$ = probability of the most likely state sequence
from 1 ... i ending in state $\pi_i = k$

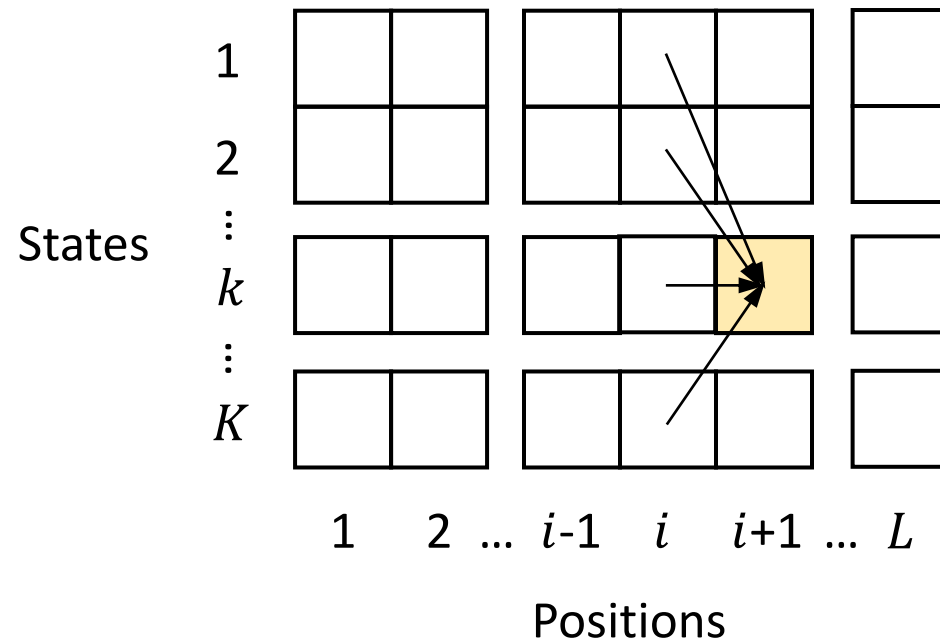
Viterbi algorithm



$$v_l(i+1) = \underbrace{e_l(x_{i+1})}_{\text{Current emission probability}} \max_k (v_k(i) a_{kl})$$

Current emission
probability

Viterbi algorithm



$$v_l(i + 1) = e_l(x_{i+1}) \underbrace{\max_k (v_k(i) a_{kl})}_{\text{Best transition from a previous state}}$$

Best transition from
a previous state

Viterbi algorithm

Algorithm: Viterbi

Initialisation ($i = 0$): $v_0(0) = 1, v_k(0) = 0$ for $k > 0$.

Recursion ($i = 1 \dots L$): $v_l(i) = e_l(x_i) \max_k (v_k(i-1) a_{kl});$
 $\text{ptr}_i(l) = \text{argmax}_k (v_k(i-1) a_{kl}).$

Termination: $P(x, \pi^*) = \max_k (v_k(L) a_{k0});$
 $\pi_L^* = \text{argmax}_k (v_k(L) a_{k0}).$

Traceback ($i = L \dots 1$): $\pi_{i-1}^* = \text{ptr}_i(\pi_i^*).$

Viterbi vs. posterior decoding

1. Viterbi algorithm:
Finds the most likely sequence of hidden states.

$$\pi^* = \operatorname{argmax}_{\pi} P(x, \pi)$$

2. Forward-backward algorithm (posterior decoding):
Finds the most likely hidden state at each position.

$$\hat{\pi}_i = \operatorname{argmax}_k \underbrace{P(\pi_i = k | x)}$$

Posterior probability of
state k at position i , given
the entire sequence

Posterior decoding

$$\hat{\pi}_i = \operatorname{argmax}_k P(\pi_i = k|x)$$

Note (chain rule):

$$P(\pi_i = k|x) = \frac{P(x, \pi_i = k)}{P(x)}$$



Probability of full sequence of observations
(sum over all possible state sequences)

$$P(x) = \sum_{\pi} P(x, \pi)$$

Posterior decoding

$$\hat{\pi}_i = \operatorname{argmax}_k P(\pi_i = k|x)$$

Note (chain rule):

$$P(\pi_i = k|x) = \frac{P(x, \pi_i = k)}{P(x)}$$

$$\begin{aligned} P(x, \pi_i = k) &= P(x_1 \dots x_i, \pi_i = k) P(x_{i+1} \dots x_L | x_1 \dots x_i, \pi_i = k) \\ &= \underbrace{P(x_1 \dots x_i, \pi_i = k)} \underbrace{P(x_{i+1} \dots x_L | \pi_i = k)} \end{aligned}$$

Both quantities can be computed recursively,
similarly to the Viterbi algorithm

Posterior decoding

$$\hat{\pi}_i = \operatorname{argmax}_k P(\pi_i = k|x)$$

Note (chain rule):

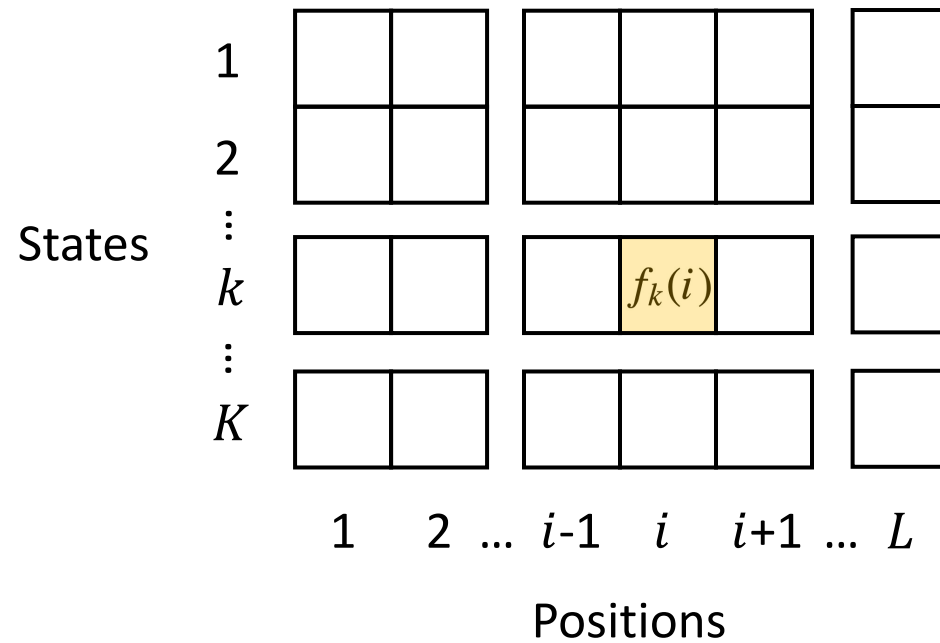
$$P(\pi_i = k|x) = \frac{P(x, \pi_i = k)}{P(x)}$$

$$\begin{aligned} P(x, \pi_i = k) &= P(x_1 \dots x_i, \pi_i = k) P(x_{i+1} \dots x_L | x_1 \dots x_i, \pi_i = k) \\ &= \underbrace{P(x_1 \dots x_i, \pi_i = k)}_{f_k(i)} \underbrace{P(x_{i+1} \dots x_L | \pi_i = k)}_{b_k(i)} \end{aligned}$$

Forward variable

Backward variable

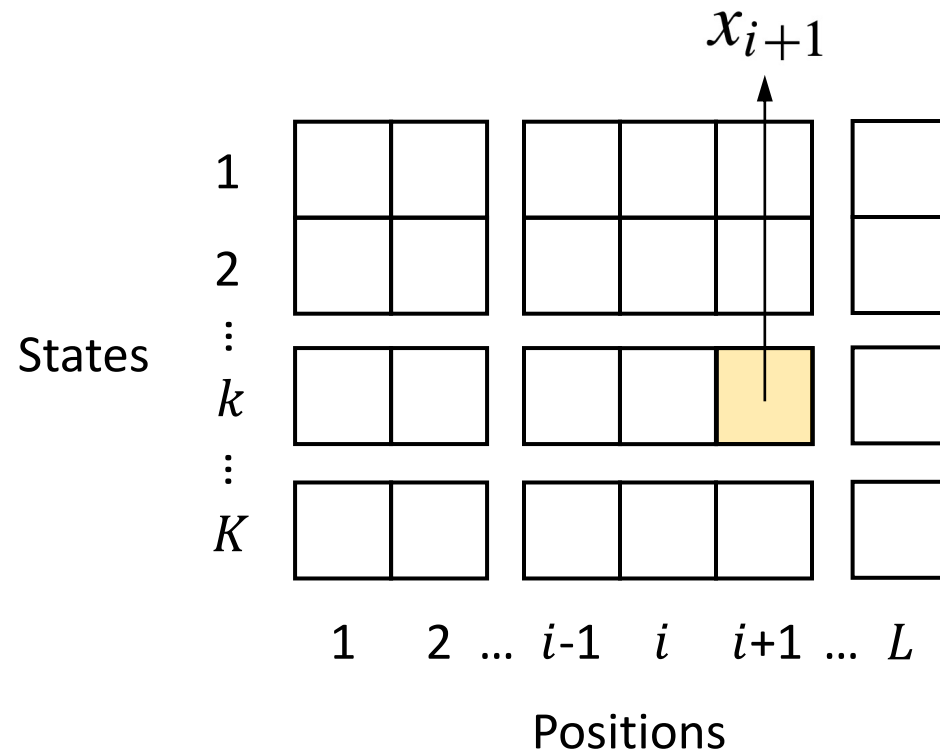
Forward algorithm



$$f_k(i) = P(x_1 \dots x_i, \pi_i = k)$$

Probability of the observed sequence from
1 ... i , with state $\pi_i = k$

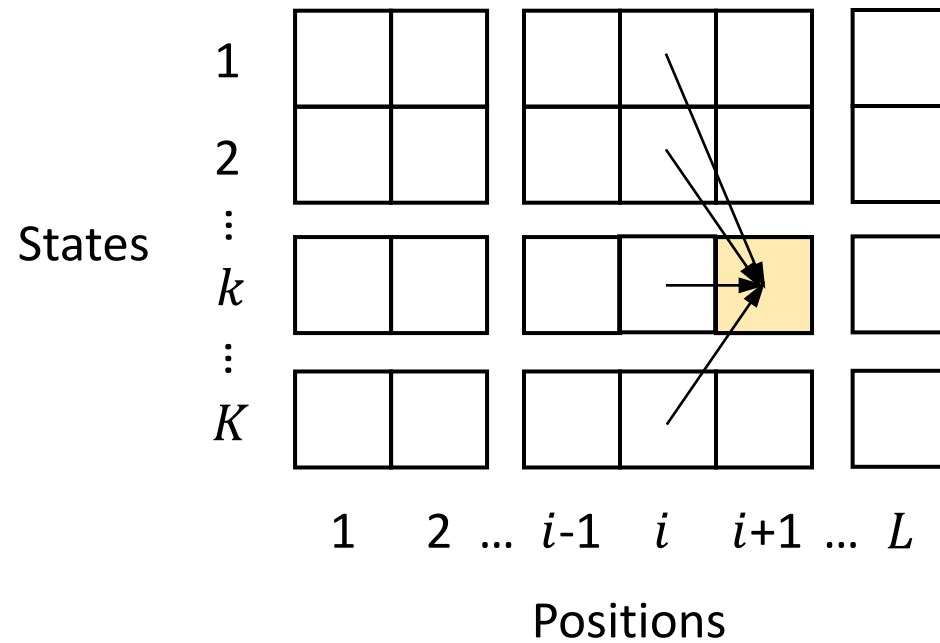
Forward algorithm



$$f_l(i+1) = \underbrace{e_l(x_{i+1})}_{\text{Current emission probability}} \sum_k f_k(i) a_{kl}$$

Current emission
probability

Forward algorithm



$$f_l(i+1) = e_l(x_{i+1}) \underbrace{\sum_k f_k(i) a_{kl}}_{\text{Sum of transitions over all previous hidden states}}$$

Sum of transitions over
all previous hidden states

Forward algorithm

Algorithm: Forward algorithm

Initialisation ($i = 0$): $f_0(0) = 1, f_k(0) = 0$ for $k > 0$.

Recursion ($i = 1 \dots L$): $f_l(i) = e_l(x_i) \sum_k f_k(i-1) a_{kl}$.

Termination: $P(x) = \sum_k f_k(L) a_{k0}$.

 Note $P(x)$ is now easily computed from the forward variables at the last position in the sequence

Posterior decoding

$$\hat{\pi}_i = \operatorname{argmax}_k P(\pi_i = k|x)$$

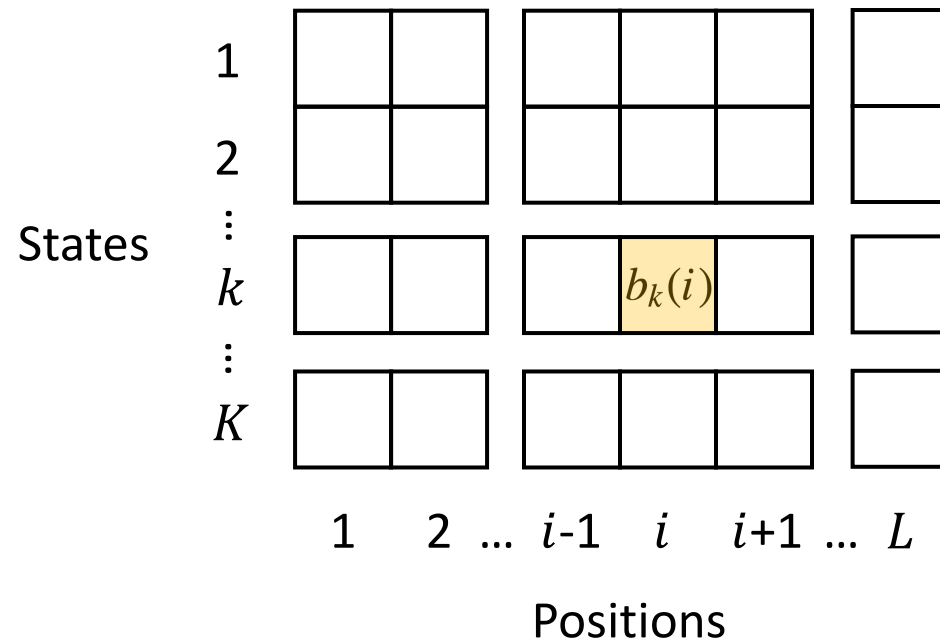
$$P(\pi_i = k|x) = \frac{P(x, \pi_i = k)}{P(x)}$$

$$\begin{aligned} P(x, \pi_i = k) &= P(x_1 \dots x_i, \pi_i = k) P(x_{i+1} \dots x_L | x_1 \dots x_i, \pi_i = k) \\ &= \underbrace{P(x_1 \dots x_i, \pi_i = k)}_{f_k(i)} \underbrace{P(x_{i+1} \dots x_L | \pi_i = k)}_{b_k(i)} \end{aligned}$$

Forward variable

Backward variable

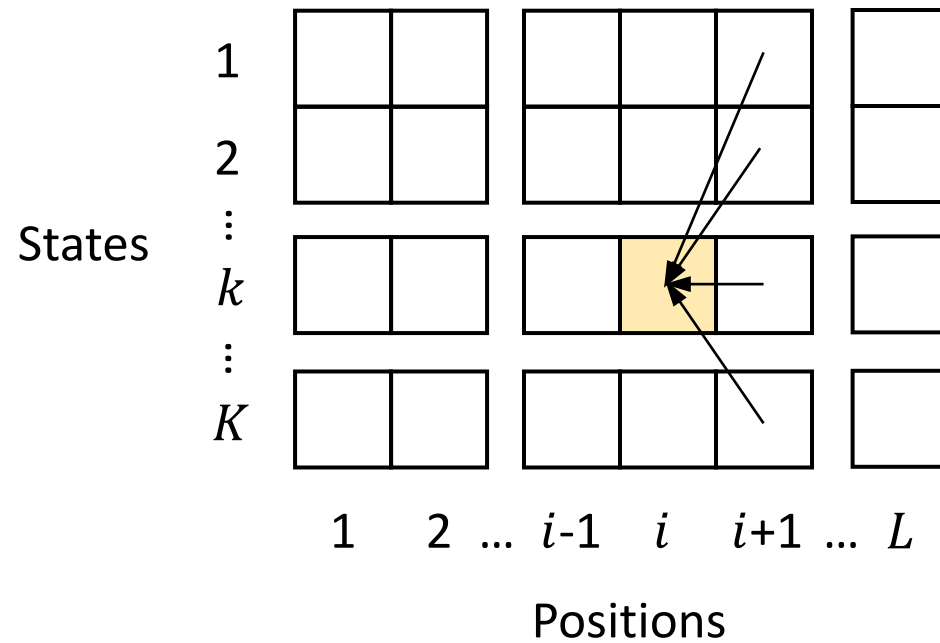
Backward algorithm



$$b_k(i) = P(x_{i+1} \dots x_L | \pi_i = k)$$

Probability of the observed sequence from
 $i+1 \dots L$, given state $\pi_i = k$

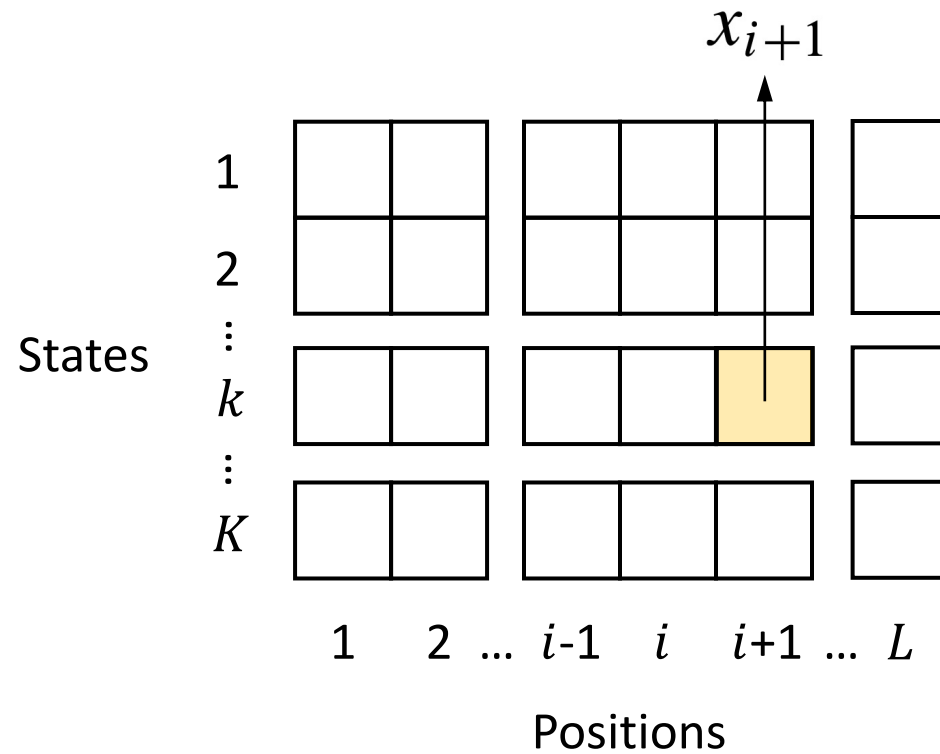
Backward algorithm



$$b_k(i) = \underbrace{\sum_l a_{kl} e_l(x_{i+1}) b_l(i+1)}_{\text{Sum over all transitions to the next hidden state}}$$

Sum over all transitions
to the next hidden state

Backward algorithm



$$b_k(i) = \sum_l a_{kl} \underbrace{e_l(x_{i+1})}_{\text{Include } i+1 \text{ emission probability (not included in } b_l(i+1))} b_l(i+1)$$

Include $i+1$ emission probability
(not included in $b_l(i+1)$)

Backward algorithm

Algorithm: Backward algorithm

Initialisation ($i = L$): $b_k(L) = a_{k0}$ for all k .

Recursion ($i = L - 1, \dots, 1$): $b_k(i) = \sum_l a_{kl} e_l(x_{i+1}) b_l(i + 1)$.

Termination: $P(x) = \sum_l a_{0l} e_l(x_1) b_l(1)$.

Posterior decoding

$$\hat{\pi}_i = \operatorname{argmax}_k P(\pi_i = k|x)$$

$$P(\pi_i = k|x) = \frac{P(x, \pi_i = k)}{P(x)}$$

$$\begin{aligned} P(x, \pi_i = k) &= P(x_1 \dots x_i, \pi_i = k) P(x_{i+1} \dots x_L | x_1 \dots x_i, \pi_i = k) \\ &= \underbrace{P(x_1 \dots x_i, \pi_i = k)}_{f_k(i)} \underbrace{P(x_{i+1} \dots x_L | \pi_i = k)}_{b_k(i)} \end{aligned}$$

Forward variable

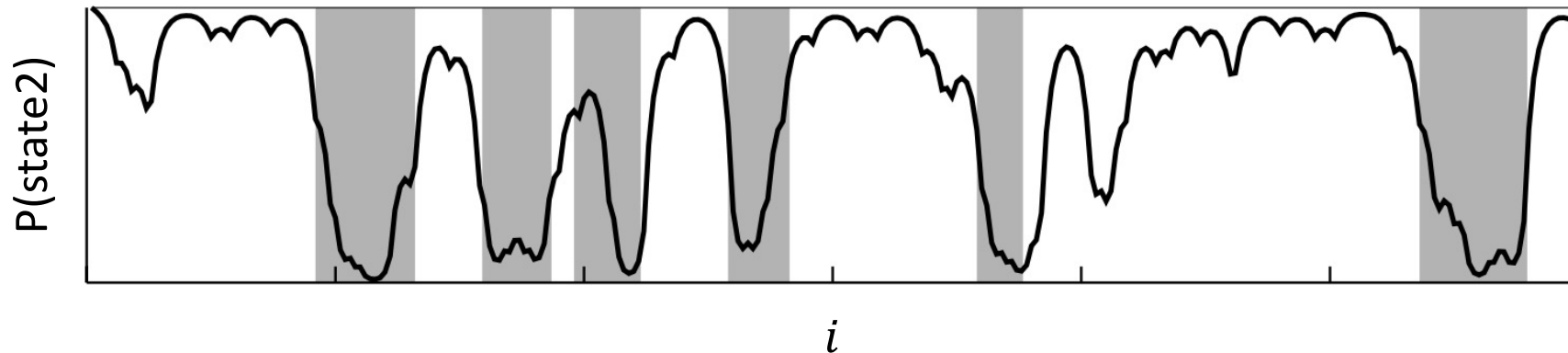
Backward variable

Posterior decoding

$$\begin{aligned}P(x, \pi_i = k) &= P(x_1 \dots x_i, \pi_i = k) P(x_{i+1} \dots x_L | \pi_i = k) \\ &= f_k(i) b_k(i)\end{aligned}$$

$$P(\pi_i = k | x) = \frac{P(x, \pi_i = k)}{P(x)} = \frac{f_k(i) b_k(i)}{P(x)}$$

e.g.:



Viterbi vs. posterior decoding

- Many hidden state sequences can have almost the same probability as the most probable one.
- If we care more about the state at a particular position i , then posterior decoding may be more appropriate.
- The sequence of most likely hidden states at each position, however, may not be a legitimate sequence if some transition probabilities are 0.

$$\hat{\pi}_i = \operatorname{argmax}_k P(\pi_i = k | x)$$